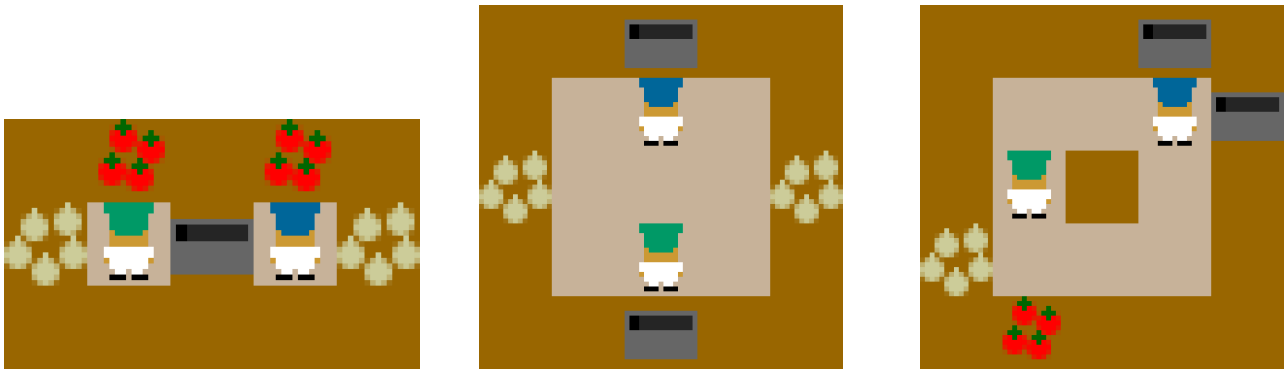


(a) **Overcooked – "Locked In":** cooks need to coordinate on the timing of a recipe using a single pot.

(b) **Overcooked – "Schelling":** cooks need to coordinate on who gets to occupy the useful central tile without bumping into each other to deposit two onions in time.

(c) **Overcooked – "Coordination Ring":** cooks cannot occupy the same spot, requiring them to coordinate on how they pass around the central island.

Figure 4: The three Overcooked environments analyzed in Section 6, each requiring different types of coordination.

The operator $\mathscr{O}(\mathcal{G}_i, h_i^t, \pi^t)$ on line 8 conceptually iterates over the class $[[\pi_{-i}]]$ of policies that are equivalent under condition (ii). It is implemented by (1) enumerating the one-step co-policies, i.e., assignments of actions to the leaves of π_{-i}^t and (2) enumerating over choices of nodes in $\mathcal{G}_i$ associated with all possible one-step extensions $h_i^t + (a_i, o_i)$ of h_i^t.

Consider computing the VSECs (Figure 3b) in state 1 of the game in Figure 3a given the intermediate VSECs of states 2 and 3. The operator $\mathscr{O}$ enumerates all choices of nodes for the two future states, and all assignments to the co-policy at state 1. For example, choosing the node $\boxed{(\{a\}, 1) : \{a\}}$ in state 2 and node $\boxed{(\{b\}, 1) : \{b\}}$ in state 3 and the one-step co-policy a produces best response $A_i^* = \{a, b\}$ in state 1.

Line 10 and 11 record the optimal actions and their value. Line 12 partitions the sets of co-policies based on their valued best response, i.e. checking condition (i). Finally, line 14 takes the computed classes and adds the corresponding node and edges to $\mathcal{G}_i$.

6. Experiments

The purpose of our experiments is twofold: (1) to characterize and gain intuition about what the strategic equivalence relation (and its equivalence classes) look like in practice, and (2) to empirically evaluate the scaling performance of our algorithms. We compute the strategic equivalence relation (SER) for different environments and provide several videos made available on our website that include 1-4 sampled representative policies from each class.

Our primary evaluation environment is the Overcooked environment (Carroll et al., 2019), in which players control chefs that cook meals in a kitchen. Agents need to coordinate on the high-level strategy for collecting the different ingredients in a dish and the low-level motion controls to avoid getting in each other's way. The onion and tomato tiles hold an infinite supply of the ingredients and agents get positive reward for placing various combinations of vegetables in the pots, depending on the specific environment. Agents have access to different movement actions (depending on the environment) and an interact action which picks up or places ingredients depending on what tile the agent is facing. We provide analysis and visualization of additional Overcooked environments in Appendix C.

6.1. Fully-observed settings

Overcooked – Locked-in. The first Overcooked environment we investigate is visualized in Figure 4a. We let the agents' action space be composed of the following: rotate left, rotate right, and interact. The agents need to collaborate to make a soup containing an onion and a tomato within 7 timesteps. However, the recipe requires placing the onion into the pot precisely one timestep after the tomato to get +1 reward, or the soup will be ruined.

This environment has two strategic equivalence classes: one class has the left agent collecting an onion and the right agent collecting a tomato, while the other class has the roles switched. Because the environment is fully-observed, both classes have both agents delaying their commitment to either vegetable at the beginning of the episode to see which vegetable their co-player will choose, and then at some point breaking the symmetry by committing to a vegetable. See the two videos on our website for visualizations.

In the rest of this section, we investigate computing valued strategic equivalence relations (VSECs) over trembling-hand subgame perfect Nash equilibrium using the optimized algorithm described in Appendix B.6.

Overcooked – Schelling Point. This Overcooked environment is depicted in Figure 4b. Agents get a reward of +1 for each onion that is put into either of the pots. We let agents also have a larger action space: move left, right, up, down, stay still, and interact. We use a horizon of $H = 8$. In this environment agents need to coordinate on who gets to occupy the useful central tile without running into each other (which results in a no-op). Optimal policies achieve a total reward of +2.

This environment has only two equivalence classes, depending on which agent gets to occupy the central square first. After one agent occupies the central square, the symmetry in the problem is broken and the second agent's best responses become fixed. See the two videos on our website for visualizations.

Overcooked – Coordination Ring. This Overcooked environment is depicted in Figure 4c. Agents get a reward of +1 as soon as there is an onion and tomato in the same pot. Again we let agents also have a larger action space: move left, right, up, down, stay still, and interact. We use a horizon of $H = 11$, although different values of H produce similar results.

At first examination, one might predict that there should be multiple VSEC in this problem, perhaps resulting from which agent gets which vegetable, which pot to use, or the various options for how the agents could walk around the central island to stay out of each other's way. However, all of these variations are found within the same strategic equivalence class!

Surprisingly, this environment has only a *single* VSEC for either agent. Conceptually, this means that both agents *do not need to know anything* about their co-policy in order to play optimally; agents require no preexisting agreements on how to successfully coordinate in the environment. Since the environment is fully-observed, the blue agent is able to watch the green agent's actions and best respond no matter what strategy green chooses. See the video on our website for a visualization of this VSEC.

6.2. Partially-observed environments

Referential Game. The first partially-observed environment we investigate is a referential game between two agents described in Figure 5. The tricky aspect of this game is that both agents need to agree on the meaning of the signal that is communicated across the channel.

We compute three natural VSECs in this game: one class where the agents agree that ■ $\mapsto$ 0 and □ $\mapsto$ 1 (which achieves optimal reward of 1), another class where the meaning is swapped, and a third class where the green agent (right) ignores the meaning of the signal and tries to guess the blue agent's (left) observation with expected value $\frac{1}{2}$.

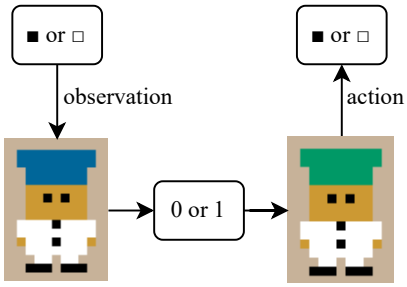


Figure 5: **Referential game:** Blue (left) first randomly observes either ■ or □. Green (right) observes a 0/1 signal from Blue and chooses ■ or □, trying to match Blue's observation.

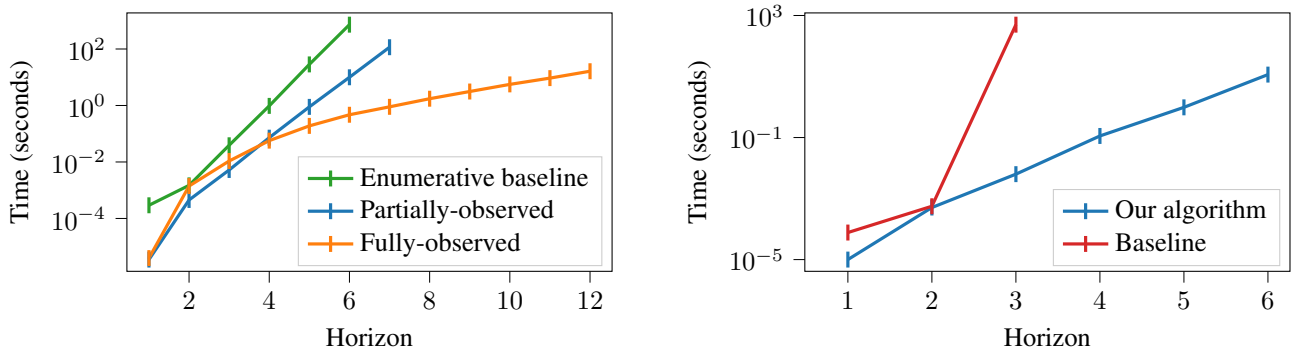
Overcooked – Locked-in. The second partially-observed environment we investigate is a modified version of the Overcooked environment depicted in Figure 4a. In this version of the environment, the central pot blocks the agents' vision, preventing them from observing the position or actions of their co-player. This modification greatly increases the degree of coordination required between the agents because they can not rely on observing and adapting to their co-player's behavior. The reward is the same as before: agents get +1 reward whenever an onion is put into the pot precisely one timestep after a tomato is put into the pot.

This greater degree of coordination is clearly reflected in the size of the strategic equivalence relation: there are six classes for either agent that vary along two axis: (1) who takes charge of which vegetable and (2) whether the tomato should enter the pot on timestep 4, 5, or, 6 (and onion one timestep after). See the six videos on our website for visualization.

6.3. Computational efficiency

Since strategic equivalence relations are a novel concept, there are no pre-existing algorithms for computing them. We compare our algorithms against an enumerative baseline. The enumerative baseline computes strategic equivalence relations by first flattening the (partially observed) DecPOMDP down to a normal-form game where actions in the normal-form game represent policies in the DecPOMDP. Then, strategic equivalence classes are computed by enumerating co-policies and grouping by best response.

We experiment on increasing the horizon of policies in the Overcooked environment depicted in Figure 4a and testing how long it takes to compute the strategic equivalence relation using different algorithms. From Figure 6a (which is on a log scale), we can see the clear benefit of exploiting recursive structure to compute strategic equivalence classes, with results amplified using the optimized algorithm for a fully-observed version of the same setting. Figure 8 in the



(a) *Fully-observed* (Algorithm in Appendix B.6) uses the fully-observed version of the environment. *Partially-observed* (Algorithm 1) and *Enumerative baseline* (described in Section 6.3) use the partially-observed version of the environment.

(b) The time it take to compute the set of best-response policies using our algorithm (Algorithm 1) and the baseline from Szer & Charpillet (2006). Beyond a horizon of two, the baseline timed-out at 30 minutes.

Figure 6: The time (on a log scale) for computing strategic equivalence relations over policies of varying horizon using varying algorithms in the Overcooked environment depicted in Figure 4a. Note that all algorithms make use of the optimization when enumerating joint policies in deterministic environments (see Appendix B.5).

Appendix shows the variant of this plot on an absolute scale.

Since our algorithm also computes the set of best-response policies, it provides notable computational complexity improvements over existing algorithms (Szer & Charpillet, 2006; Seuken & Zilberstein, 2007) for computing the best-response policies in DecPOMDPs. By grouping co-policies into VSECs, the backup step of our algorithm avoids redundant computation from computing the best-responses to co-policies that are in the same SEC. Therefore, the number of multiagent beliefs that we need to consider for each history no longer scales in the exponential space of our co-policies, but rather in the number of intermediate VSECs over those co-policies. Figure 6b shows the time it take to compute the set of best-response policies using our algorithm (Algorithm 1 and *Partially-observed* in Figure 6a) and the baseline from Szer & Charpillet (2006). All algorithms are implemented in Python using the same data structures and use the optimization in deterministic environments (see Appendix B.5).

7. Discussion

We have shown that the distinction between strategically relevant and irrelevant information can be formalized via the novel concept of strategic equivalence relation (SER). By providing a compact representation of this relation, and an efficient Bellman backup to compute it, we have provided an approach to efficiently understand what is strategically relevant to any given task. This allows us to shed new light on existing coordination benchmarks. For instance, we show in Section 6 that there is non-trivial strategically relevant information in Overcooked only in situations in which simultaneous decisions between incompatible optimal joint plans must be made. Given that such situations are rare, this provides a theoretical explanation as to why relatively good coordination with humans is achievable in this domain without any human data (Strouse et al., 2021). Our framework also explains why the introduction of simultaneous decisions (most easily through partial observability) can lead to benchmarks that are specifically more challenging for coordination, such as Hanabi (Bard et al., 2020).

The SER could serve as a critical component of both centralized and decentralized approaches for creating cooperative policies. In a decentralized settings, the SER tells us exactly what is needed in order to optimally coordinate with co-players. This could allow a focused effort on few-shot or zero-shot adaptation to co-players (Zand et al., 2022; Albrecht & Stone, 2017; Stone et al., 2010; Hu et al., 2020), or tell us how to prioritize strategically relevant information if only a limited communication bandwidth is available (Wang et al., 2020; Mao et al., 2020; Berna-Koes et al., 2004).

In a centralized setting, the SER tells us what must be agreed upon before the problem may be reduced to independent single-agent problems. As such, it can provide critical information for finding useful decompositions of centralized value functions (Jin et al., 2022; Cassano et al., 2021; Wang* et al., 2020), or more efficient DecPOMDP planning algorithms (Szer & Charpillet, 2006; Seuken & Zilberstein, 2007). In fact, in Section 6.3, we showed that our algorithm for computing the SER already represents a complexity improvement in the state-of-the-art for computing the set of best-response strategies.

While we are conscious of the computational challenges to scale our approach to complex domains, we are optimistic about future work to approximate our method. In this work,